

11.1-11.2

we perturb from nice things - zeeman and hyperfine splitting

$$|\varphi_n^{(1)}\rangle = \sum_k |\varphi_k^{(0)}\rangle \langle \varphi_k^{(0)} | \varphi_n^{(1)} \rangle = |\varphi_n^{(0)}\rangle \langle \varphi_n^{(0)} | \varphi_n^{(1)} \rangle + \sum_{k \neq n} |\varphi_k^{(0)}\rangle \langle \varphi_k^{(0)} | \varphi_n^{(1)} \rangle \quad (11.18)$$

$$|\psi_n\rangle = |\varphi_n^{(0)}\rangle + \lambda \sum_{k \neq n} |\varphi_k^{(0)}\rangle \frac{\langle \varphi_k^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} + \mathcal{O}(\lambda^2) \quad (11.23)$$

We look at first and second order approximations to the perturbation

11.2

degeneracy bad

$$\sum_{i=1}^N \langle \varphi_{n,j}^{(0)} | \hat{H}_1 | \varphi_{n,i}^{(0)} \rangle c_i = E_n^{(1)} \sum_{i=1}^N \langle \varphi_{n,j}^{(0)} | \varphi_{n,i}^{(0)} \rangle c_i = E_n^{(1)} \sum_{i=1}^N \delta_{ji} c_i \quad (11.31)$$

We assume that degenerate states are orthonormal, and make a basis set

$$\begin{pmatrix} (H_1)_{11} & (H_1)_{12} \\ (H_1)_{21} & (H_1)_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E_n^{(1)} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad (11.34)$$